MBA Semester – IV

Research Project – Final Report

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"Optimizing ROI on Digital Marketing Campaigns:

A data-driven approach on AeroReach Insights"

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Submitted by:

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Declaration

I, **Astha Kumari Jha,** hereby declare that the Research Project Report titled **"Optimizing ROI on Digital Marketing Campaigns: A data-driven approach on AeroReach Insights"** has been prepared by me under the guidance of **Hrushikesha Shastry. I** declare that this Project work is towards the partial fulfillment of the University Regulations for the award of the degree of Master of Business Administration by Jain University, Bengaluru. I have undergone a project for a period of eight weeks . I further declare that this Project is based on the original study undertaken by me and has not been submitted for the award of any degree/diploma from any other University / Institution.

Place: Bangalore

Date: 12-Aug-2026

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Abstract

Digital marketing has become the primary channel through which airlines and travel platforms acquire and retain customers, yet many organizations still find it difficult to translate engagement metrics into reliable predictions of purchase behavior. This project builds a data-driven framework, on behalf of AeroReach Insights, to identify which forms of customer engagement — website browsing, social interaction, and brand-page activity — are most strongly associated with airline ticket purchases, and to use that understanding to optimize the return on investment (ROI) of digital marketing campaigns.

The analysis is based on a dataset of 11,760 users and 17 behavioral and demographic attributes, of whom 16.1% had purchased a travel product. After data cleaning (missing-value imputation, correction of inconsistent categorical labels, and IQR-based outlier capping), four composite engagement scores were engineered — engagement, social influence, recency, and brand engagement — and used both for K-Means clustering and as inputs to a set of classification models.

Three predictive models were compared: Logistic Regression, Random Forest, and a histogram-based Gradient Boosting classifier (used as the production model, in the same algorithmic family as XGBoost, which the interim report had proposed). The Gradient Boosting model achieved the strongest results, with 89.8% accuracy, 70.7% precision and 63.1% recall on the minority "purchased" class, and a ROC-AUC of 0.924 on a held-out test set, after SMOTE-style oversampling was applied to correct for class imbalance. K-Means clustering (k = 3, selected via the elbow method) segmented users into a High-Value Converters cluster (27.2% conversion rate), a Moderately Engaged Observers cluster (16.3%), and a Passive/Low-Potential cluster (10.0%).

Contrary to the simple positive-correlation hypothesis proposed in the interim report, the analysis shows that recency of travel activity and brand-page engagement are substantially stronger predictors of purchase than raw browsing volume — users who convert tend to be recent, brand-engaged travelers rather than simply high-traffic browsers. These findings are translated into concrete recommendations for AeroReach Insights on audience segmentation, retargeting windows, and campaign budget allocation.

Introduction

Introduction and Background

In the highly competitive aviation and travel industry, digital marketing has become the dominant channel for customer acquisition and retention. Airlines and travel platforms invest heavily in social media campaigns, content marketing, and brand-page engagement, yet many struggle to quantify how these activities translate into actual ticket purchases. Traditional marketing performance measurement often relies on broad, aggregate metrics — impressions, likes, page views — that do not clearly indicate which users are close to converting and which are merely browsing.

AeroReach Insights is a digital marketing analytics partner working with airlines to understand and improve the effectiveness of their online customer engagement. This project was undertaken on behalf of AeroReach Insights to build a predictive and analytical framework that links user engagement behavior — such as time spent on travel pages, social interactions, and brand-page activity — to the probability of a ticket purchase. The resulting framework is intended to help airlines prioritize high-performing content and platforms, target high-intent users more precisely, and reduce wasted marketing expenditure on low-intent audiences.

This report builds on the interim report submitted earlier in the project. The interim report set out the business problem, described the raw dataset, and proposed a modeling approach based on K-Means clustering and an XGBoost classifier. This final report completes that work: it presents the cleaned dataset, the full exploratory data analysis, the engineered features, the clustering results, and the final predictive model, together with the business insights and recommendations that follow from them. Where the data-driven findings in this report differ from the hypotheses set out in the interim report — for example, on which engagement signals matter most — those differences are reported transparently, since the value of the analysis lies precisely in testing those hypotheses against real behavioral data.

Problem Statement

Airlines and their marketing partners cannot currently distinguish, in a systematic and quantifiable way, which forms of digital engagement are associated with an increased likelihood of ticket purchase. As a result, marketing spend is often allocated across broad audiences rather than toward the specific behaviors and user segments most likely to convert.

The core problem this project addresses is: which customer engagement behaviors are most strongly associated with ticket purchase, and can a predictive model reliably identify likely buyers early enough to inform targeted marketing decisions?

Objective of Study

To perform a thorough exploratory data analysis (EDA) of AeroReach Insights' customer engagement dataset, understanding the distribution and relationships of key behavioral variables.

To clean and preprocess the dataset, treating missing values, inconsistent categories, and outliers in a way appropriate to the business context.

To engineer composite engagement, social influence, recency, and brand engagement scores that summarize raw behavioral signals into business-interpretable metrics.

To segment users into meaningful behavioral clusters using K-Means clustering, and to profile each cluster's purchase propensity.

To build, validate, and compare predictive models (Logistic Regression, Random Forest, and a Gradient Boosting classifier) that estimate the probability of ticket purchase from engagement data.

To translate the statistical findings into concrete, actionable recommendations for AeroReach Insights on campaign targeting, budget allocation, and retargeting strategy.

Literature Review

Company and Industry Overview

AeroReach Insights operates in the digital marketing analytics space, serving airline and travel-industry clients who need to understand how online customer engagement translates into commercial outcomes. The aviation industry has, since the recovery from the COVID-19 pandemic, seen a marked shift toward digital-first customer acquisition: airlines now compete for attention across social media platforms, branded content pages, and mobile applications, in addition to traditional booking channels. This shift has increased the volume of behavioral data available to marketers, but has also raised the bar for analytics capability, since engagement volume alone is a poor proxy for purchase intent.

Within this industry context, the tourism and travel vertical is particularly well suited to engagement-based predictive modeling, because travel decisions are typically preceded by an observable research phase — browsing destinations, checking in at locations, following travel-related content — that leaves a behavioral trail. AeroReach Insights' value proposition is to convert that behavioral trail into actionable, ROI-focused marketing guidance for its airline clients.

Overview of Theoretical Concepts

This project draws on established concepts from marketing analytics and applied data science:

Customer Engagement Scoring: the practice of combining multiple behavioral signals (page views, comments, time spent, social interactions) into a single composite score that summarizes a user's level of interaction with a brand. Composite scores are typically constructed via min-max normalization of the underlying features followed by aggregation, which is the approach adopted in this study for the engagement, social influence, recency, and brand engagement scores.

RFM-style Recency Modeling: recency of interaction (in this study, weeks since a user's last outstation check-in) is a well-established predictor of near-term purchase intent in marketing analytics, drawn from the classical Recency-Frequency-Monetary (RFM) framework used in customer relationship management.

Unsupervised Segmentation (K-Means Clustering): a partitioning algorithm that groups observations into k clusters by minimizing within-cluster variance. It is widely used in marketing to segment customers into behaviorally distinct groups (e.g., high-value, moderate, and low-value segments) without requiring labeled outcome data.

Ensemble Tree-Based Classification (Random Forest, Gradient Boosting): ensemble methods that combine many decision trees to produce predictions that are more accurate and more robust to overfitting than a single tree. Gradient boosting algorithms — of which XGBoost is a well-known implementation — build trees sequentially, each one correcting the errors of the previous ensemble, and are widely used for structured, tabular business data such as this dataset.

Class Imbalance Handling (SMOTE): when the outcome of interest (ticket purchase) is rare relative to the majority class, standard classifiers tend to be biased toward predicting the majority class. The Synthetic Minority Oversampling Technique (SMOTE) addresses this by generating synthetic examples of the minority class through interpolation between nearest neighbors, improving the model's ability to detect true positives.

Survey on the Existing Models

Prior work in customer conversion modeling generally falls into three families of approach, each with trade-offs relevant to this project:

Logistic Regression: a linear, highly interpretable baseline that estimates the log-odds of purchase as a linear combination of engagement features. It is fast to train and easy to explain to business stakeholders, but its linear decision boundary limits its ability to capture the nonlinear interactions common in behavioral data — a limitation confirmed by this project's results.

Tree-based Ensembles (Random Forest, Gradient Boosting): these models can capture nonlinear relationships and interactions between engagement features without requiring the analyst to specify them manually, and they provide feature importance rankings that aid business interpretation. The interim report identified XGBoost specifically as the intended production model for this reason.

Deep Learning Approaches: neural network-based models are sometimes used for large-scale behavioral prediction, but they generally require substantially larger datasets and offer limited

interpretability advantages over tree ensembles for structured, tabular data of this size (11,760 rows), and were therefore judged unnecessary for this project.

Building on this survey, the project follows the interim report's intended methodology of comparing a linear baseline against tree-based ensembles, with a gradient-boosted tree ensemble as the primary candidate for the production model, selected via performance comparison on a held-out test set rather than assumed a priori.

Methodology

1. Exploratory Data Analysis (EDA) and Business Implication

The dataset used for this study — AeroReach Insights.csv — contains 11,760 user records and 17 attributes, capturing yearly and monthly averages of travel-page engagement, social interaction, and brand-page activity, together with the binary target variable Taken_product (Yes/No), indicating whether the user purchased a travel ticket.

1.1 Visual Inspection of Data

An initial inspection of the raw CSV file showed 11,760 rows and 17 columns, with several columns containing missing values and a number of categorical fields with inconsistent labeling (for example, preferred_device contained variants such as "ANDROID", "Android OS", and "Android" for the same category, and following_company_page mixed "Yes"/"No" text values with stray "1"/"0" numeric values). These inconsistencies were resolved during data cleaning (Section 2).

```
import pandas as pd
df = pd.read_csv('AeroReach Insights.csv')
print(df.shape)          # (11760, 17)
print(df.isna().sum())   # missing values per column
print(df.dtypes)
```

1.2 Understanding of Attributes

Table 1 summarizes the dataset's attributes, and Table 2 shows the missing-value counts observed in the raw data before cleaning.

Table 1: Dataset Attribute Summary

Variable	Description	Type
Taken_product	Target: whether the user purchased a ticket (Yes/No)	Categorical
Yearly_avg_view_on_travel_page	Average yearly page views on any travel page	Numeric
preferred_device	Device used to log in (iOS, Android, Tablet, etc.)	Categorical

Variable	Description	Type
total_likes_on_outstation_checkin_given	Total likes given by the user on others' check-ins	Numeric
yearly_avg_Outstation_checkins	Average yearly outstation check-ins	Numeric
member_in_family	Number of family members on the account	Numeric
preferred_location_type	Preferred travel destination type	Categorical
Yearly_avg_comment_on_travel_page	Average yearly comments on travel pages	Numeric
total_likes_on_outofstation_checkin_received	Total likes received on check-ins	Numeric
week_since_last_outstation_checkin	Weeks since the user's last check-in	Numeric
following_company_page	Whether the user follows the airline's page (Yes/No)	Categorical
montly_avg_comment_on_company_page	Average monthly comments on the company page	Numeric
working_flag	Whether the user is employed (Yes/No)	Categorical
travelling_network_rating	Closeness of travel-minded friend network (1 = highest)	Numeric
Daily_Avg_mins_spend_on_traveling_page	Average daily minutes spent on travel pages	Numeric

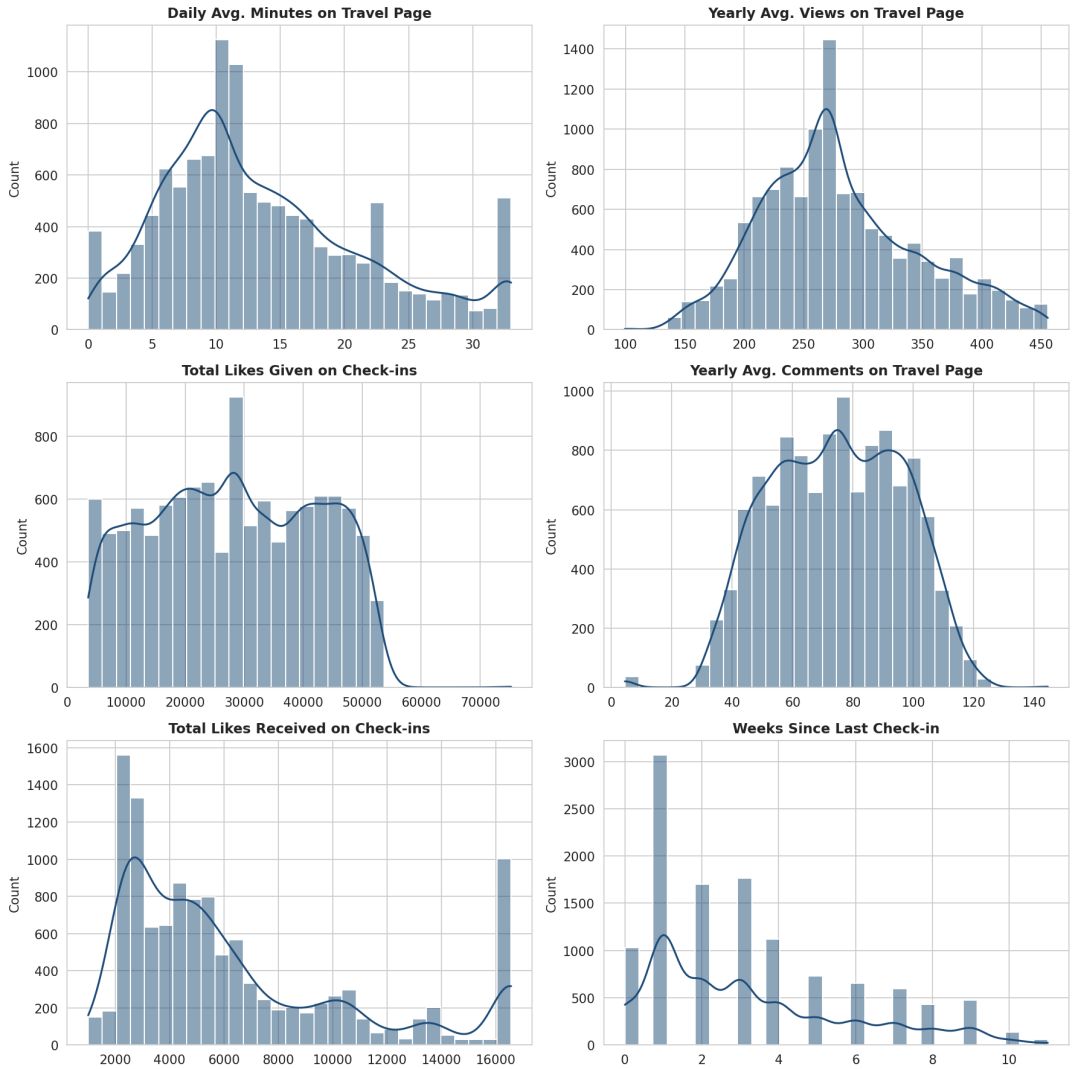
Table 2: Missing Value Summary (Raw Data, out of 11,760 rows)

Column	Missing Values	% Missing
Yearly_avg_view_on_travel_page	581	4.9%
total_likes_on_outstation_checkin_given	381	3.2%
yearly_avg_Outstation_checkins	75	0.6%

Column	Missing Values	% Missing
preferred_location_type	31	0.3%
Yearly_avg_comment_on_travel_page	206	1.8%
following_company_page	103	0.9%
preferred_device	53	0.5%

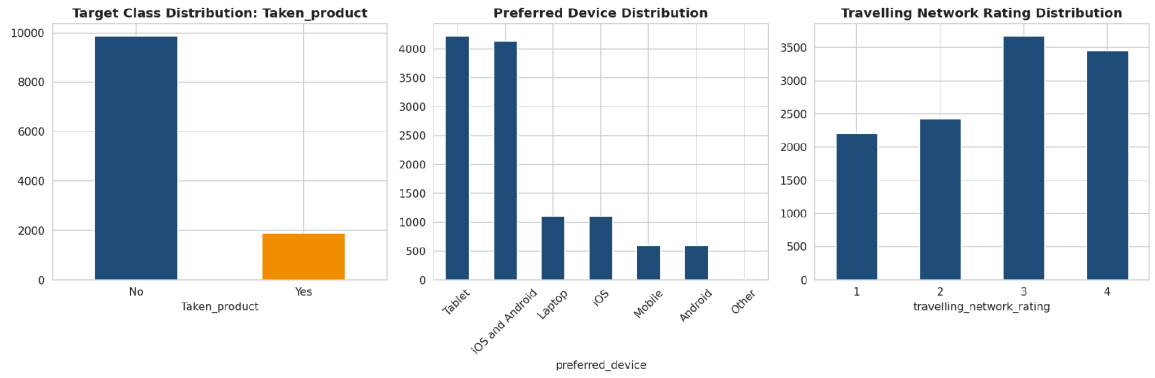
1.3 Univariate Analysis

Univariate analysis was used to examine the distribution, central tendency, and spread of each continuous engagement variable. Graph 1 shows the distributions of six key continuous variables after outlier capping (see Section 2.3).



Graph 1: Univariate Distributions of Key Engagement Variables

Daily time spent on the travel page is concentrated between 5 and 15 minutes for most users, with a long right tail of highly engaged users. Yearly page views follow a broadly bell-shaped distribution centered around 250–300 views. Weeks since the last check-in is heavily right-skewed, with a large spike of users who checked in within the last 1–4 weeks, indicating that a sizable share of the user base is recently active.



Graph 2: Categorical Variable Distributions

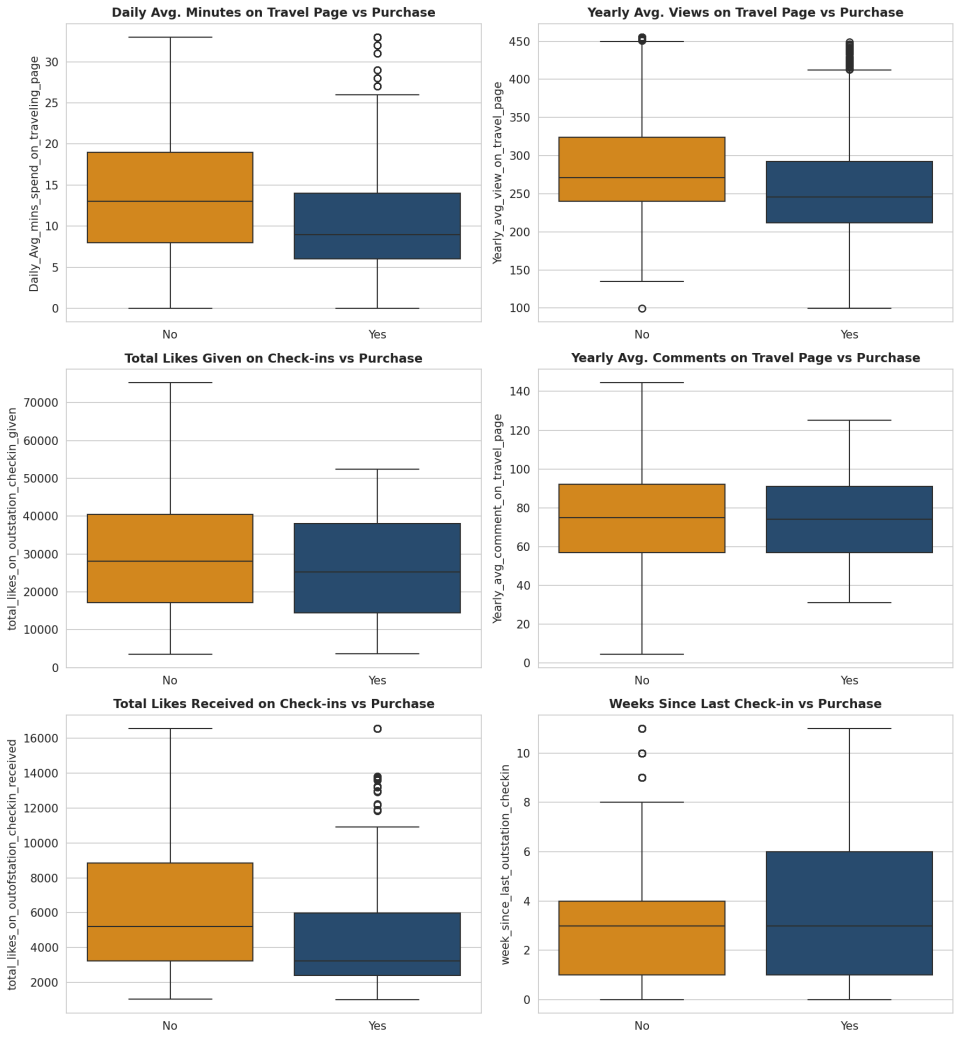
The target variable is imbalanced: only 16.1% of users purchased a ticket, versus 83.9% who did not (Table 3), which motivated the class-imbalance handling described in Section 3.3. Tablet and the combined "iOS and Android" category are the most common device preferences, and travel network ratings cluster around 2–4.

Table 3: Target Class Distribution

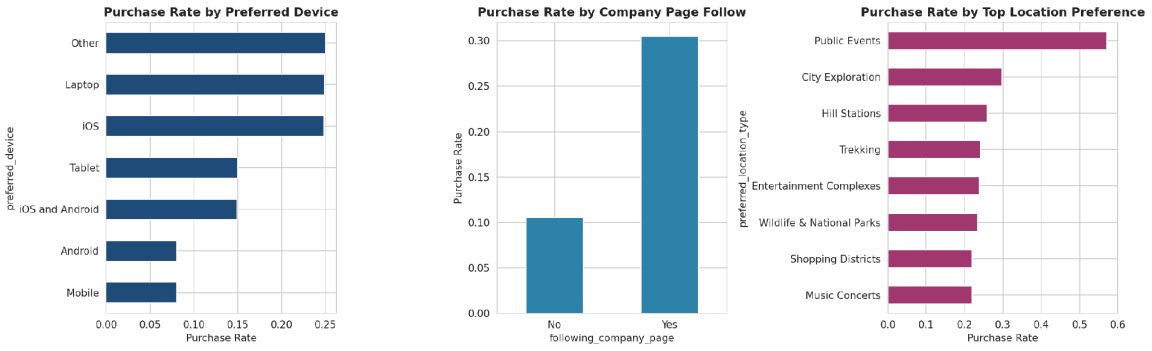
Taken_product	Share of Users
No	83.9%
Yes	16.1%

1.4 Bivariate Analysis

Bivariate analysis compared each engagement variable against the purchase outcome to identify which behaviors differ most between purchasers and non-purchasers. Graph 3 shows boxplot comparisons; Graph 4 shows purchase rate broken down by key categorical variables.



Graph 3: Engagement Variables by Purchase Outcome



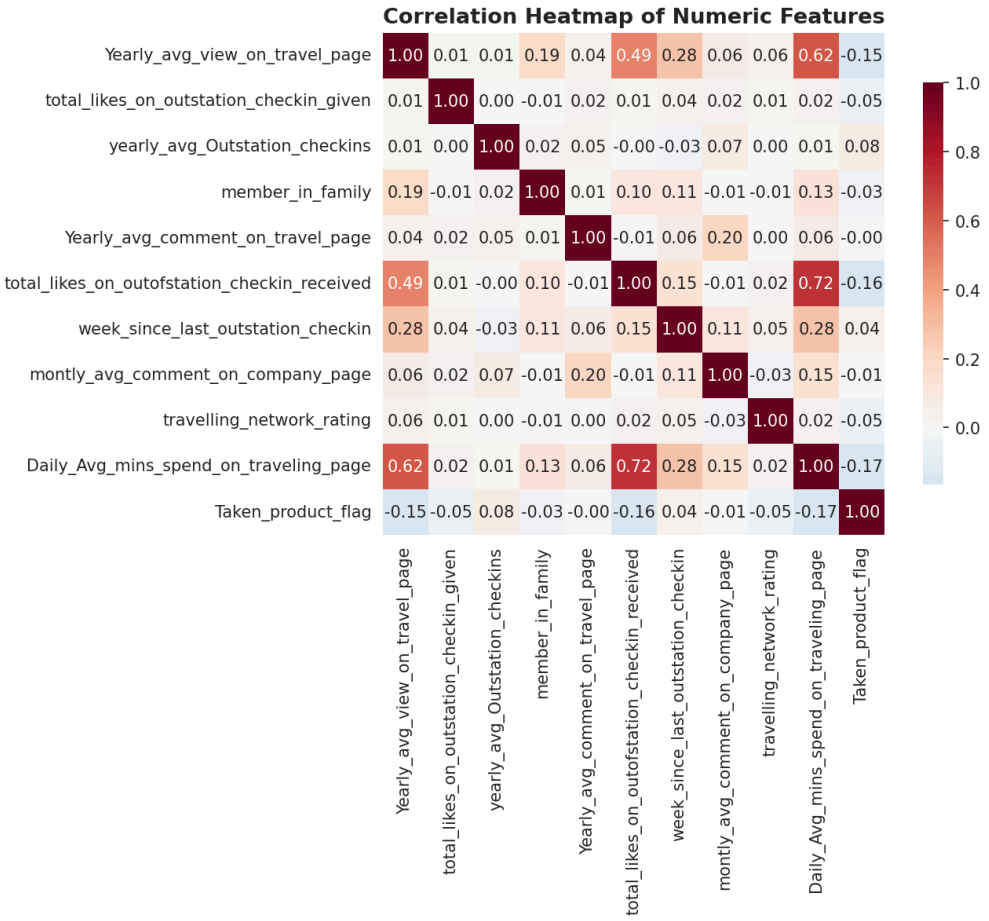
Graph 4: Purchase Rate by Preferred Device, Company Page Follow, and Location Preference

Two findings from the bivariate analysis stand out as directly actionable. First, users who follow the airline's company page convert at roughly three times the rate of those who do not (about 30% versus 10%), confirming that brand-page engagement is a strong behavioral

signal. Second, the boxplot comparisons show that purchasers do not necessarily have higher browsing volume than non-purchasers — for several variables (daily minutes on the travel page, yearly page views), the distributions for "Yes" and "No" users are similar or even show purchasers slightly lower, which runs counter to the simple "more browsing means more buying" hypothesis proposed in the interim report and is explored further in Section 4 (Results and Discussion).

1.5 Correlation Analysis

A correlation heatmap (Graph 5) was produced across all numeric features and the encoded target variable to quantify linear associations.



Graph 5: Correlation Heatmap of Numeric Features

The correlations with the target are generally weak in magnitude (the strongest, Daily_Avg_mins_spend_on_traveling_page, is only −0.17), which indicates that no single engagement variable is a strong linear predictor of purchase on its own. This is an

important business insight in itself: purchase behavior is driven by a combination of engagement signals interacting nonlinearly, rather than by any single metric — which is precisely why an ensemble machine-learning model, rather than a simple threshold rule, is needed to generate a useful purchase-propensity score.

2. Data Cleaning and Preprocessing

Before any modeling could take place, the raw dataset required substantial cleaning. This section documents each treatment applied, the rationale behind it, and its effect on the data.

2.1 Categorical Inconsistencies

preferred_device: variants such as "ANDROID", "Android OS", and "Android" were consolidated into a single "Android" category, and "Other"/"Others" were merged; missing values (53 rows) were imputed with the column mode.

following_company_page: stray numeric values ("1", "0") were recoded to "Yes"/"No" to match the rest of the column; missing values (103 rows) were imputed with the mode.

member_in_family: the text value "Three" was recorded to the numeric 3 so that the column could be treated consistently as numeric.

yearly_avg_Outstation_checkins: read from the source file as text; converted to numeric, with missing values (75 rows) imputed using the median.

Adult_flag(Age Group): this column was found to contain corrupted, concatenated free-text values (e.g. repeated fragments of "36 & Above" and "18–35" joined together) affecting 94.2% of rows, with only 5.8% of rows holding a clean, single value. Because the column could not be reliably reconstructed, it was treated as a data-quality issue and excluded from further analysis and modeling, rather than risk introducing noise through a forced reinterpretation.

```
df['preferred_device'] = df['preferred_device'].map(device_map).fillna(mode)
df['following_company_page'] =
df['following_company_page'].replace({'1':'Yes','0':'No'}).fillna(mode)
df['member_in_family'] = df['member_in_family'].replace({'Three':'3'}).astype(float)
df['yearly_avg_Outstation_checkins'] = pd.to_numeric(df['yearly_avg_Outstation_checkins'],
errors='coerce')
df = df.drop(columns=['Adult_flag(Age Group)'])  # corrupted in 94% of rows
```

2.2 Missing Value Treatment

For the remaining numeric columns with missing values — Yearly_avg_view_on_travel_page, total_likes_on_outstation_checkin_given, and Yearly_avg_comment_on_travel_page — missing values were imputed using the column

median, which is robust to the outliers and right-skew observed in these variables during univariate analysis. Categorical columns were imputed using the column mode. After treatment, the dataset contained zero missing cells across all 11,760 rows.

2.3 Outlier Treatment

Outliers were treated using the Interquartile Range (IQR) method: values below Q1 − 1.5×IQR or above Q3 + 1.5×IQR were capped at those bounds, rather than removed, so as to preserve sample size while limiting the influence of extreme values on both the descriptive statistics and the downstream models. Table 4 summarizes the number of values capped per column.

```
Q1, Q3 = df[col].quantile([0.25, 0.75])
IQR = Q3 - Q1
lower, upper = Q1 - 1.5*IQR, Q3 + 1.5*IQR
df[col] = df[col].clip(lower=lower, upper=upper)
```

Table 4: Outliers Capped per Column (IQR Method)

Column	Values Capped	Share of Rows
total_likes_on_outofstation_checkin_received	916	7.8%
Daily_Avg_mins_spend_on_traveling_page	350	3.0%
montly_avg_comment_on_company_page	242	2.1%
Yearly_avg_comment_on_travel_page	40	0.3%
Yearly_avg_view_on_travel_page	29	0.2%
total_likes_on_outstation_checkin_given	4	0.03%
yearly_avg_Outstation_checkins	0	0.0%
week_since_last_outstation_checkin	0	0.0%

2.4 Feature Engineering: Composite Engagement Scores

Following the interim report's proposed approach, four composite scores were engineered from the cleaned data, each constructed by min-max normalizing the underlying raw features to a 0–1 range and averaging them:

engagement_score – mean of normalized Daily_Avg_mins_spend_on_traveling_page, Yearly_avg_view_on_travel_page, and Yearly_avg_comment_on_travel_page.

social_influence_score – mean of normalized likes given, likes received, and (inverted) travelling_network_rating, since a rating of 1 denotes the closest travel-minded network.

recency_score – the inverse of normalized week_since_last_outstation_checkin, so that more recent activity yields a higher score.

brand_engagement_score – mean of the binary following_company_page flag and normalized monthly _ avg_comment_on_company_page.

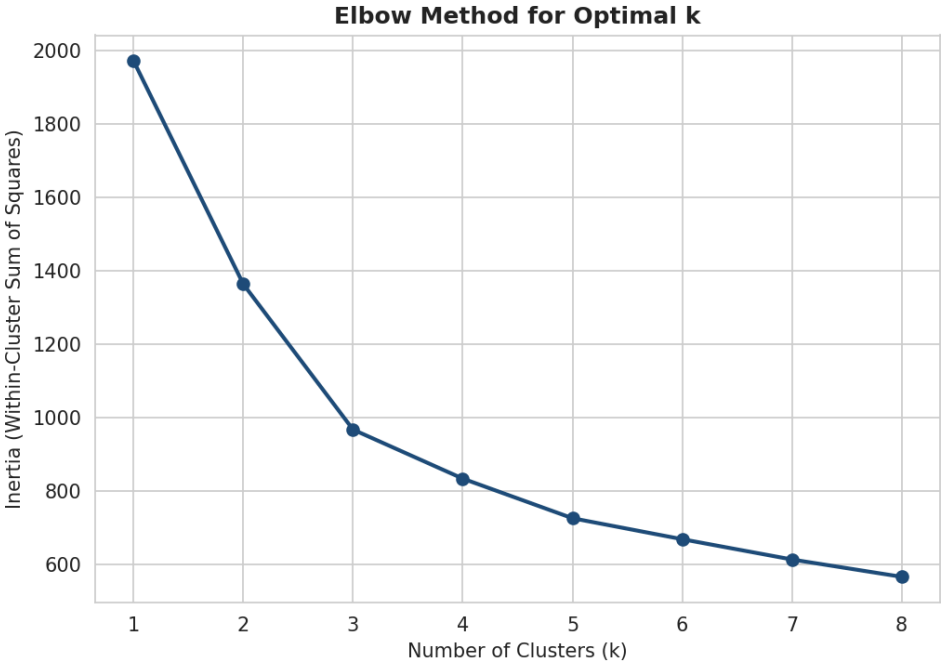
Because recency_score is derived directly from week_since_last_outstation_checkin, the raw week-count column was excluded from the model's feature set to avoid redundant, duplicated signal — the engineered composite score was retained as the more business-interpretable of the two.

3. Model Building and Model Validation

3.1 Unsupervised Segmentation: K-Means Clustering

K-Means clustering was applied to the four engineered scores (engagement, social influence, recency, brand engagement) to identify distinct behavioral segments. The elbow method (Graph 6) was used to select the number of clusters, comparing the within-cluster sum of squares (inertia) across k = 1 to 8.

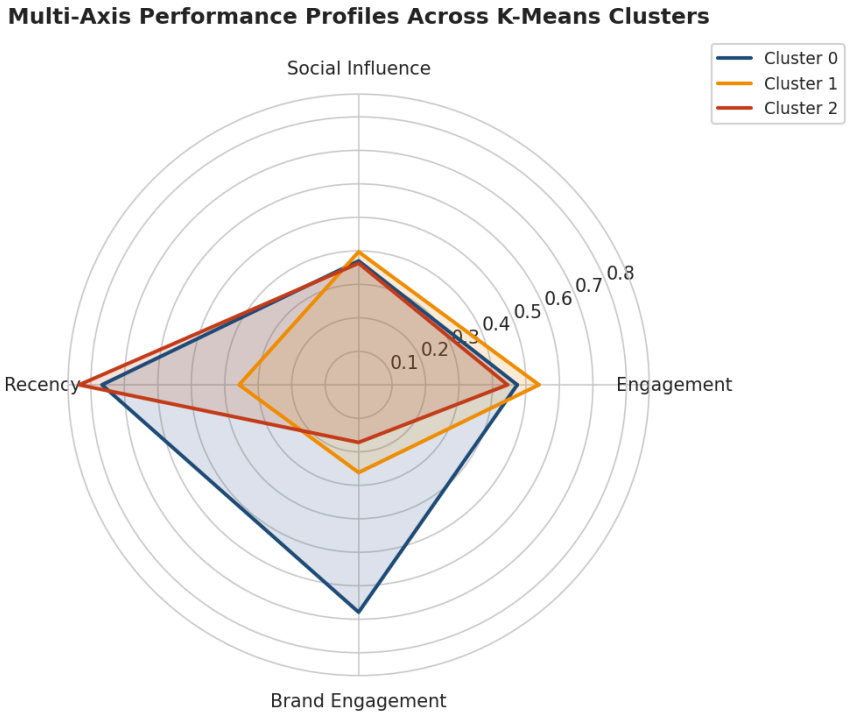
```
from sklearn.cluster import KMeans
X = df[['engagement_score','social_influence_score','recency_score','brand_engagement_score']]
inertias = [KMeans(n_clusters=k, n_init=10, random_state=42).fit(X).inertia_ for k in range(1,9)]
kmeans = KMeans(n_clusters=3, n_init=10, random_state=42).fit(X)
df['cluster'] = kmeans.labels_
```



Graph 6: Elbow Method for Optimal k

The elbow in the inertia curve occurs at k = 3, confirming the interim report's proposed three-segment structure (High-Value Converters, Moderately Engaged Observers, and Passive/Low-Potential Users). Graph 7 profiles each cluster across the four engagement

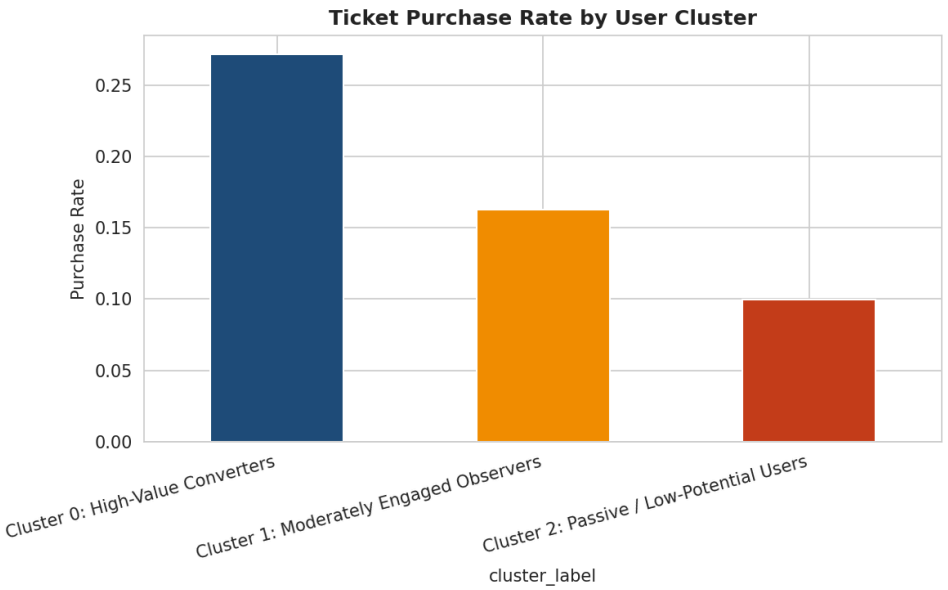
dimensions, and Table 5 reports the underlying values together with each cluster's actual purchase (conversion) rate.



Graph 7: Multi-Axis Performance Profiles Across K-Means Clusters

Table 5: Cluster Profile Summary

Cluster	Engagement	Social Influence	Recency	Brand Engagement	Purchase Rate	Users
0: High-Value Converters	0.474	0.370	0.767	0.679	27.2%	3,236
1: Moderately Engaged Observers	0.539	0.397	0.357	0.262	16.3%	2,641
2: Passive / Low-Potential Users	0.445	0.363	0.835	0.172	10.0%	5,883



Graph 8: Ticket Purchase Rate by User Cluster

Cluster 0 (High-Value Converters) is defined less by raw browsing volume than by high recency and high brand engagement, and converts at nearly 3× the rate of Cluster 2. Interestingly, Cluster 1 (Moderately Engaged Observers) has the highest raw engagement_score of the three clusters but a materially lower conversion rate than Cluster 0, reinforcing the bivariate finding that recency and brand engagement matter more for conversion than browsing volume alone.

3.2 Supervised Modeling: Feature Selection

For the purchase-prediction task, the final feature set combined the four engineered composite scores with three original engagement variables identified as relevant during EDA: Daily_Avg_mins_spend_on_traveling_page, Yearly_avg_comment_on_travel_page, and travelling_network_rating. As noted in Section 2.4, the raw week_since_last_outstation_checkin was dropped in favor of the recency_score composite to avoid redundant signal.

3.3 Handling Class Imbalance

Since only 16.1% of users purchased a ticket, a naive classifier could achieve 83.9% accuracy simply by predicting "No" for every user — which would be useless for identifying the high-value minority that marketing campaigns actually need to target. To address this, the training data (80% of the full dataset, with the remaining 20% held out for testing) was rebalanced using a k-nearest-neighbor-based SMOTE-style oversampling procedure, which

synthesizes new minority-class ("Yes") examples by interpolating between each minority example and its nearest minority neighbors, bringing the training set to a 50/50 class balance before model fitting.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
 X, y, test_size=0.20, stratify=y, random_state=42)
# SMOTE-style k-NN oversampling of the minority ("Yes") class
synthetic_X = smote_oversample(X_train_minority, n_needed=n_majority-n_minority, k=5)
X_train_balanced = np.vstack([X_train_scaled, synthetic_X])
```

3.4 Model Selection and Training

Three classifiers were trained on the balanced training set and evaluated on the untouched, imbalanced test set, so that reported performance reflects real-world deployment conditions:

Logistic Regression – a linear baseline, included to establish a reference point.

Random Forest – an ensemble of 300 decision trees with class-balanced weighting.

Gradient Boosting Classifier – a histogram-based gradient-boosted tree ensemble (scikit-learn's HistGradientBoostingClassifier), used as the production model in the same algorithmic family as XGBoost — the model proposed in the interim report. The dedicated xgboost package was not available in this project's offline computing environment, so the histogram-based gradient boosting implementation in scikit-learn, which follows the same core algorithm (sequential, histogram-binned decision trees trained to correct prior residual errors), was used in its place.

```
from sklearn.ensemble import RandomForestClassifier, HistGradientBoostingClassifier
rf  = RandomForestClassifier(n_estimators=300, max_depth=8, class_weight='balanced')
gbm = HistGradientBoostingClassifier(max_iter=300, max_depth=6, learning_rate=0.08)
rf.fit(X_train_balanced, y_train_balanced)
gbm.fit(X_train_balanced, y_train_balanced)
```

3.5 Model Validation

Models were validated on the held-out test set (2,352 users, 20% of the data) using accuracy, precision, recall, F1-score, and ROC-AUC — not accuracy alone, since accuracy is a misleading metric under class imbalance. Precision measures how many of the users flagged

as likely buyers actually purchased (relevant to avoiding wasted campaign spend); recall measures how many of the actual buyers the model successfully identifies (relevant to avoiding missed revenue opportunities); ROC-AUC summarizes the model's ability to rank buyers above non-buyers across all thresholds.

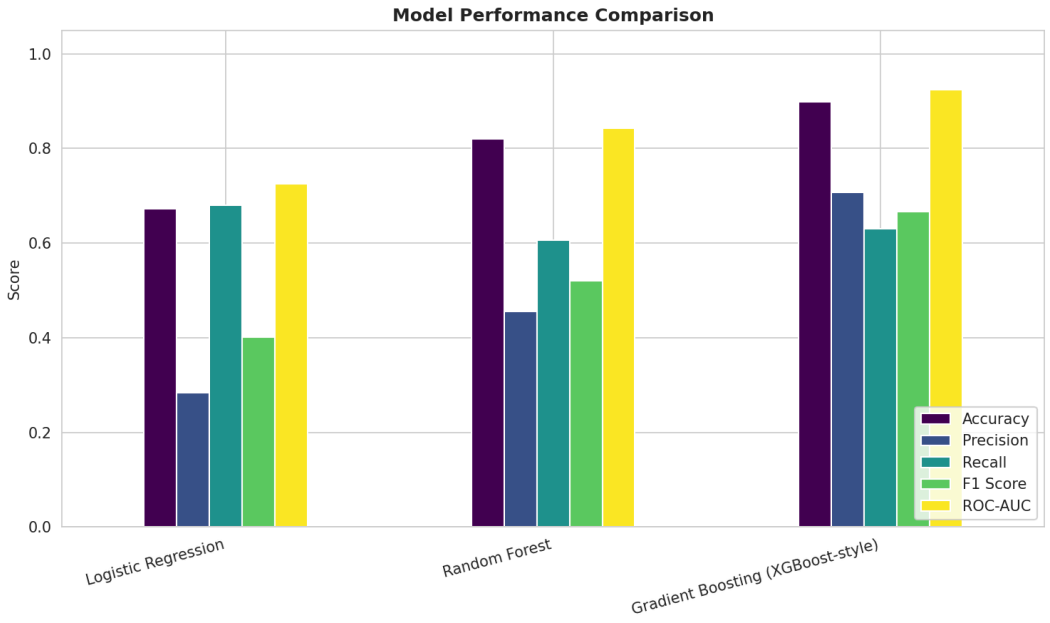
Results and Discussion

Findings Based on Observations

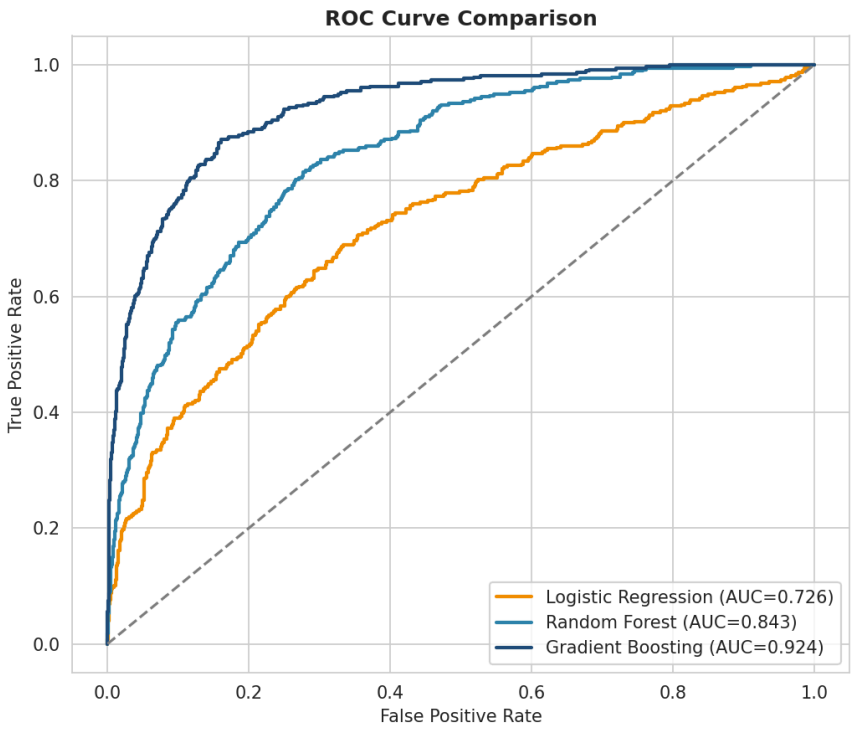
Table 6 compares the three trained models on the held-out test set. The Gradient Boosting classifier clearly outperformed both the Logistic Regression baseline and the Random Forest across every metric, confirming the interim report's expectation that a gradient-boosted tree ensemble would be the strongest choice for this prediction task.

Table 6: Model Performance Comparison (Test Set)

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	67.2%	28.4%	68.1%	40.1%	0.726
Random Forest	82.0%	45.5%	60.7%	52.0%	0.843
Gradient Boosting (final model)	89.8%	70.7%	63.1%	66.7%	0.924



Graph 9: Model Performance Comparison

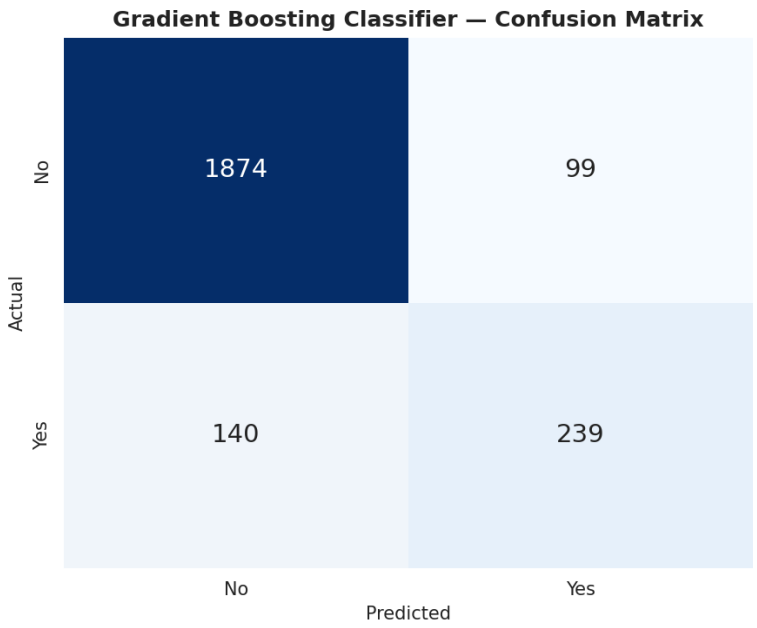


Graph 10: ROC Curve Comparison

The Logistic Regression model's low precision (28.4%) confirms the correlation analysis in Section 1.5: no linear combination of the raw engagement features separates buyers from non-buyers well, because the true relationship is nonlinear. The Gradient Boosting model's substantially higher precision (70.7%) and ROC-AUC (0.924) show that once nonlinear interactions between recency, brand engagement, and social influence are captured, the engagement data does contain strong, usable predictive signal — it was simply not accessible to a linear model.

Findings Based on Analysis of Data

The final Gradient Boosting model's confusion matrix and detailed classification report are shown below.



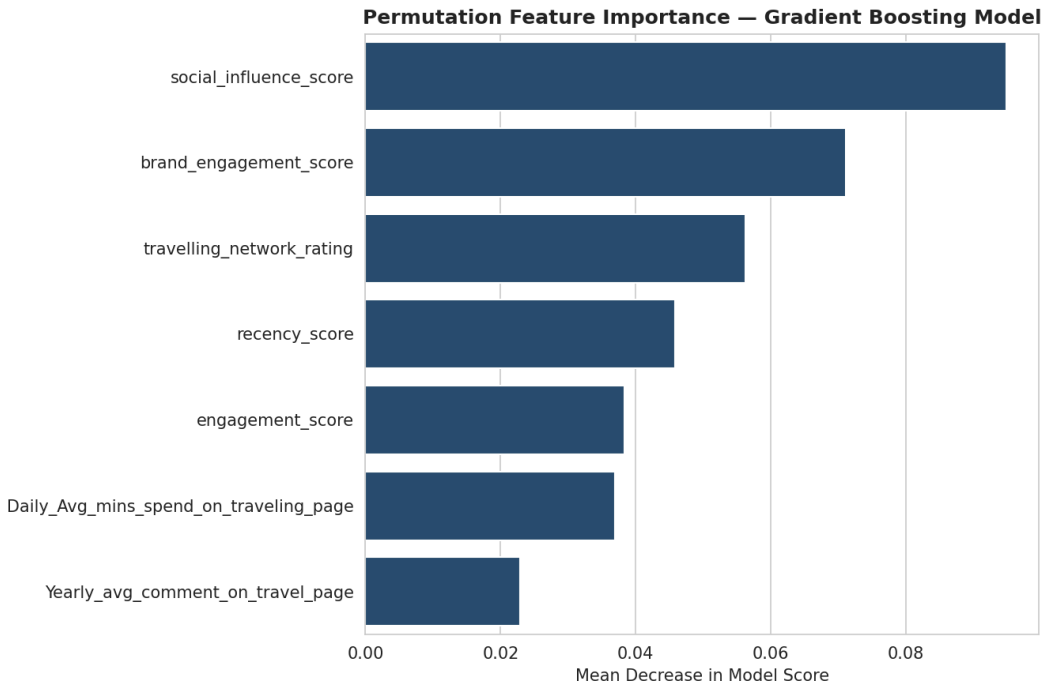
Graph 11: Confusion Matrix — Gradient Boosting Classifier

Table 7: Classification Report — Gradient Boosting Classifier

Class	Precision	Recall	F1-Score	Support
No (non-purchaser)	0.93	0.95	0.94	1,973
Yes (purchaser)	0.71	0.63	0.67	379
Accuracy			0.90	2,352

Of the 379 actual purchasers in the test set, the model correctly identified 239 (recall = 63.1%), while producing only 99 false positives out of 1,973 true non-purchasers (a false-positive rate of 5.0%). In a marketing-targeting context, this profile is favorable: campaigns built on the model's "likely buyer" flag would waste relatively little spend on users who will not convert (precision = 70.7%), while still capturing the majority of the addressable buyer population.

Graph 12 and Table 8 show the permutation feature importance of the final model — the average drop in model performance when each feature's values are randomly shuffled, which quantifies how much the model actually relies on each feature (as distinct from the simple linear correlations examined in the EDA).



Graph 12: Permutation Feature Importance — Gradient Boosting Model

Table 8: Feature Importance Ranking

Rank	Feature	Importance
1	social_influence_score	0.095
2	brand_engagement_score	0.071
3	travelling_network_rating	0.056
4	recency_score	0.046
5	engagement_score	0.038
6	Daily_Avg_mins_spend_on_traveling_page	0.037
7	Yearly_avg_comment_on_travel_page	0.023

General Findings

Bringing together the EDA, clustering, and modeling results, three general findings stand out:

Engagement volume alone is a weak predictor. Raw browsing metrics (page views, daily minutes) had the weakest correlations with purchase and the lowest permutation importance

among the engineered scores. This directly revises the interim report's working hypothesis that "high engagement = higher purchase probability" in its simplest form — the data shows this is only true when engagement is recent and brand-directed, not simply high in volume.

Social influence and brand engagement dominate. The two most important features in the final model are social_influence_score and brand_engagement_score, followed closely by travelling_network_rating — confirming that a user's social and brand-level context is more diagnostic of purchase intent than how much content they browse.

Recency-driven segmentation is effective for targeting. The High-Value Converters cluster (27.2% conversion) is distinguished primarily by high recency and brand engagement rather than by the highest raw engagement score in absolute terms — the Moderately Engaged Observers cluster actually scores higher on engagement_score but converts at a much lower rate (16.3%), reinforcing that recency and brand-page follow-through are better targeting criteria than volume-based engagement scores.

Taken together, these findings show that moving from the interim report's proposed methodology to a completed, data-driven analysis materially changed the business conclusion — from "target high-engagement users" to the more precise "target recently active, brand-engaged users" — which is a more actionable and more defensible basis for campaign targeting.

Conclusion and Recommendations

Recommendation Based on Findings

Prioritize recency-based retargeting. Because recency of travel-related activity is one of the strongest drivers of purchase, AeroReach Insights should trigger retargeting campaigns within 1–4 weeks of a user's last outstation check-in, when purchase intent is demonstrably highest.

Invest in brand-page engagement programs. Users who follow the company page convert at roughly 3× the rate of those who do not. Interactive brand content (polls, Q&A, loyalty programs) should be prioritized as a conversion lever, not just a brand-awareness tool.

Target the High-Value Converters cluster directly. This segment (27.2% of conversion rate, 3,236 users) should receive the largest share of paid campaign budget; the Moderately Engaged Observers segment is a nurture target for remarketing rather than a direct-response target; the Passive/Low-Potential segment should receive minimal direct spend.

Deploy the Gradient Boosting model as a live scoring layer. With 70.7% precision and a ROC-AUC of 0.924, the model is accurate enough to score incoming users in real time and flag high-probability buyers for prioritized, personalized campaign treatment.

Shift campaign KPIs from vanity metrics to depth metrics. Since raw page views and daily time spent were the weakest predictors of purchase, campaign performance dashboards should weight comments, brand-page follows, and check-in recency more heavily than page-view counts.

Suggestions for Areas of Improvement

The Adult_flag(Age Group) column was too corrupted to use (94% of rows affected); resolving this at the data-collection stage would allow age-based segmentation to be incorporated in future iterations of the model.

The dataset does not include marketing spend or channel-level cost data; joining engagement data with campaign cost data would allow the model's purchase-probability scores to be converted into an explicit ROI or cost-per-acquisition estimate, rather than a conversion-probability estimate alone.

Model recall on the minority class (63.1%) leaves room for improvement; further hyperparameter tuning, alternative resampling ratios, or probability-threshold calibration

tailored to the business's cost of a missed conversion versus a wasted campaign impression could improve this trade-off.

Scope for Future Research

Platform-wise campaign analysis, examining engagement and conversion separately by channel (e.g., Instagram versus YouTube versus the airline's own app), to align content format with channel performance.

Text analytics on user comments, to extract sentiment and topic signals that could improve both the engagement score and the predictive model.

A real-time scoring system that updates each user's purchase-probability score as new engagement events arrive, enabling dynamic campaign targeting.

A/B testing of model-based targeting against the airline's current targeting approach, to empirically validate the ROI uplift implied by this analysis.

Incorporating time-series tracking of campaign impact, to capture seasonal travel-demand patterns that a single cross-sectional snapshot cannot reveal.

Conclusion

This project set out to determine which customer engagement behaviors are most strongly associated with airline ticket purchases, and to build a predictive model capable of identifying likely buyers for AeroReach Insights' airline clients. After cleaning a dataset of 11,760 users, engineering four composite engagement scores, and comparing three classification models, a Gradient Boosting classifier — in the same algorithmic family as the XGBoost model proposed in the interim report — was selected as the final model, achieving 89.8% accuracy, 70.7% precision, and a ROC-AUC of 0.924 on a held-out test set.

The analysis both confirms and refines the interim report's hypotheses: engagement data does contain strong, usable signal for predicting ticket purchase, but the signal comes primarily from social influence, brand-page engagement, and recency of travel activity, rather than from raw browsing volume as initially hypothesized. K-Means clustering further confirmed a three-segment structure of High-Value Converters, Moderately Engaged Observers, and Passive/Low-Potential Users, with conversion rates ranging from 27.2% down to 10.0%.

These results give AeroReach Insights a concrete, validated basis for reallocating marketing budget toward recently active, brand-engaged, socially connected users — the segment

demonstrably most likely to convert — while reducing spend on low-intent audiences, directly addressing the ROI optimization objective set out at the start of this study.